

Molecular Physiology of Vascular Endothelial Cells

Principles and Practices of Osteopathic Medicine II

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Learning Objectives

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Learning Objective 1: Identify the factors released by endothelial cells and its effects on vascular function

Learning Objective 2: Describe the main role and differences between endothelial and vascular smooth muscle cells

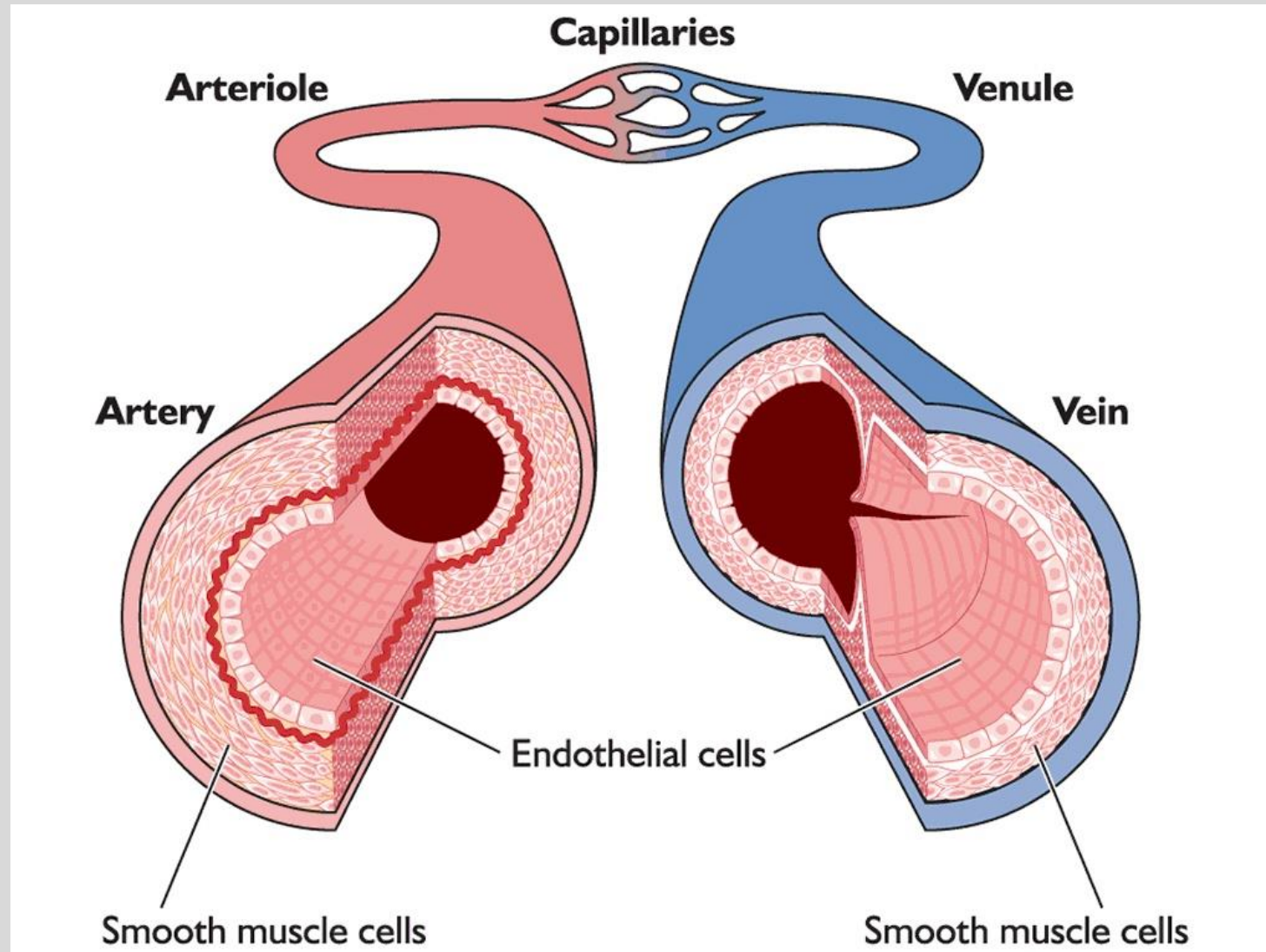
Learning Objective 3: Summarize the main signaling pathway involved in the release of nitric oxide from endothelial cells.

Learning Objective 4: Describe endothelial dysfunction and its associated complications

Vascular Structure

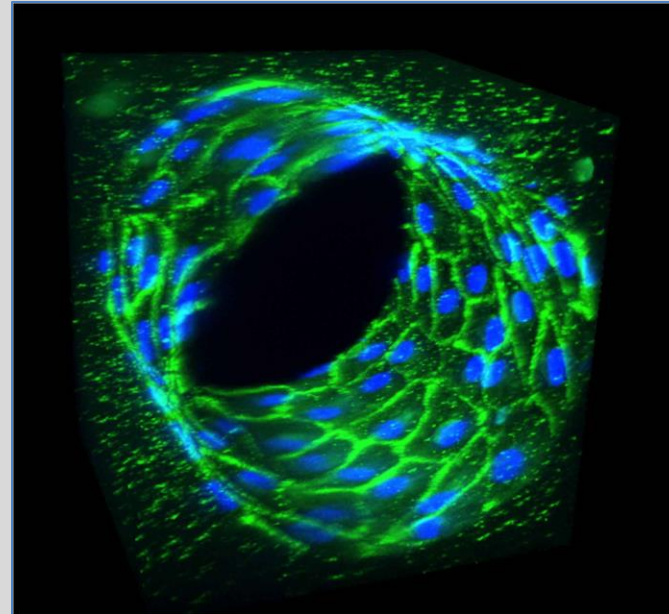
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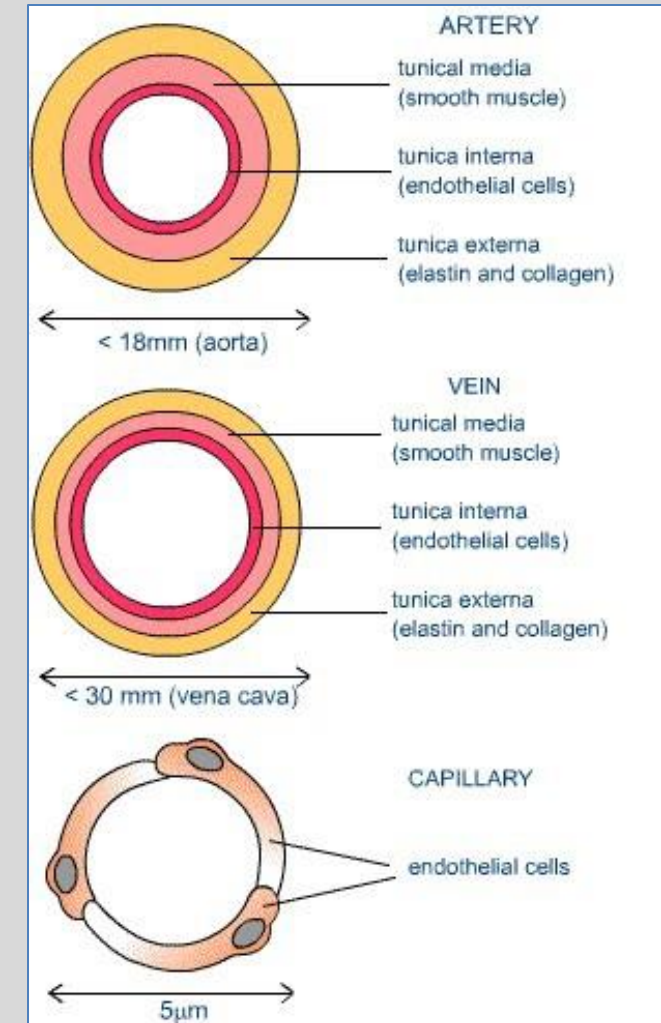
Vascular Structure

- Vessel layers
 - **Tunica interna (endothelium)**
 - Tunica media (smooth muscle)
 - Tunica externa (elastin/collagen)
- Arteries have thicker smooth muscle layers than veins (think about their function in relation to the structure)
- Endothelium is **monolayer** (smooth muscle is multilayer)
 - Green=endothelial cells
 - Blue=nuclei
- **Endothelium is a semipermeable membrane** to allow for exchange of nutrients and oxygen especially through the capillaries



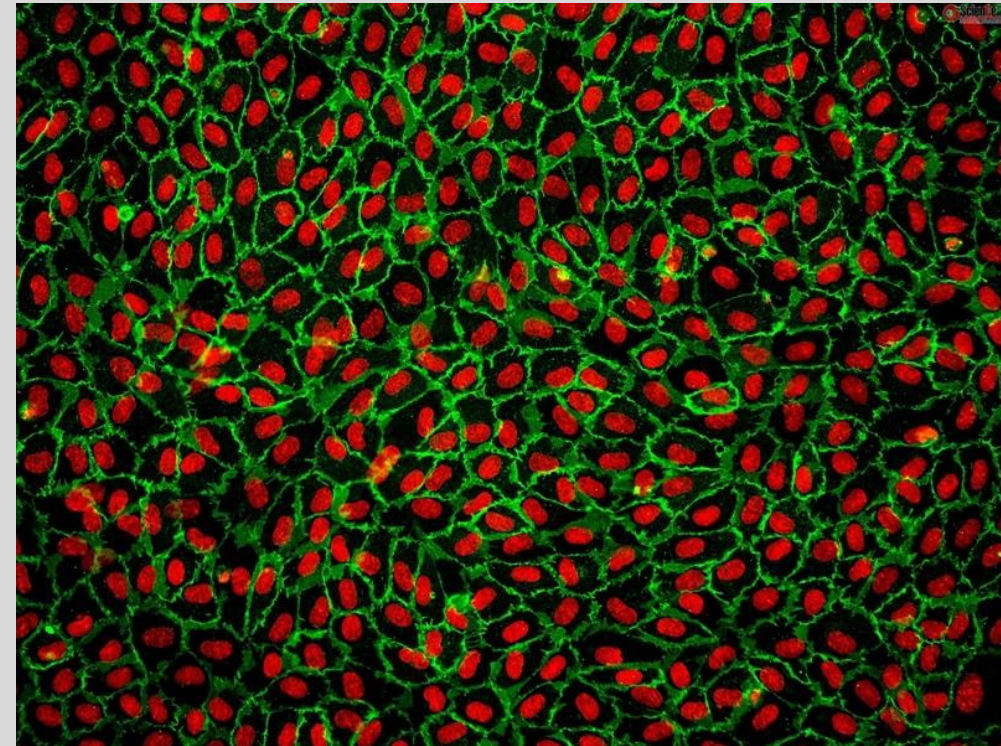
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What are these cells?

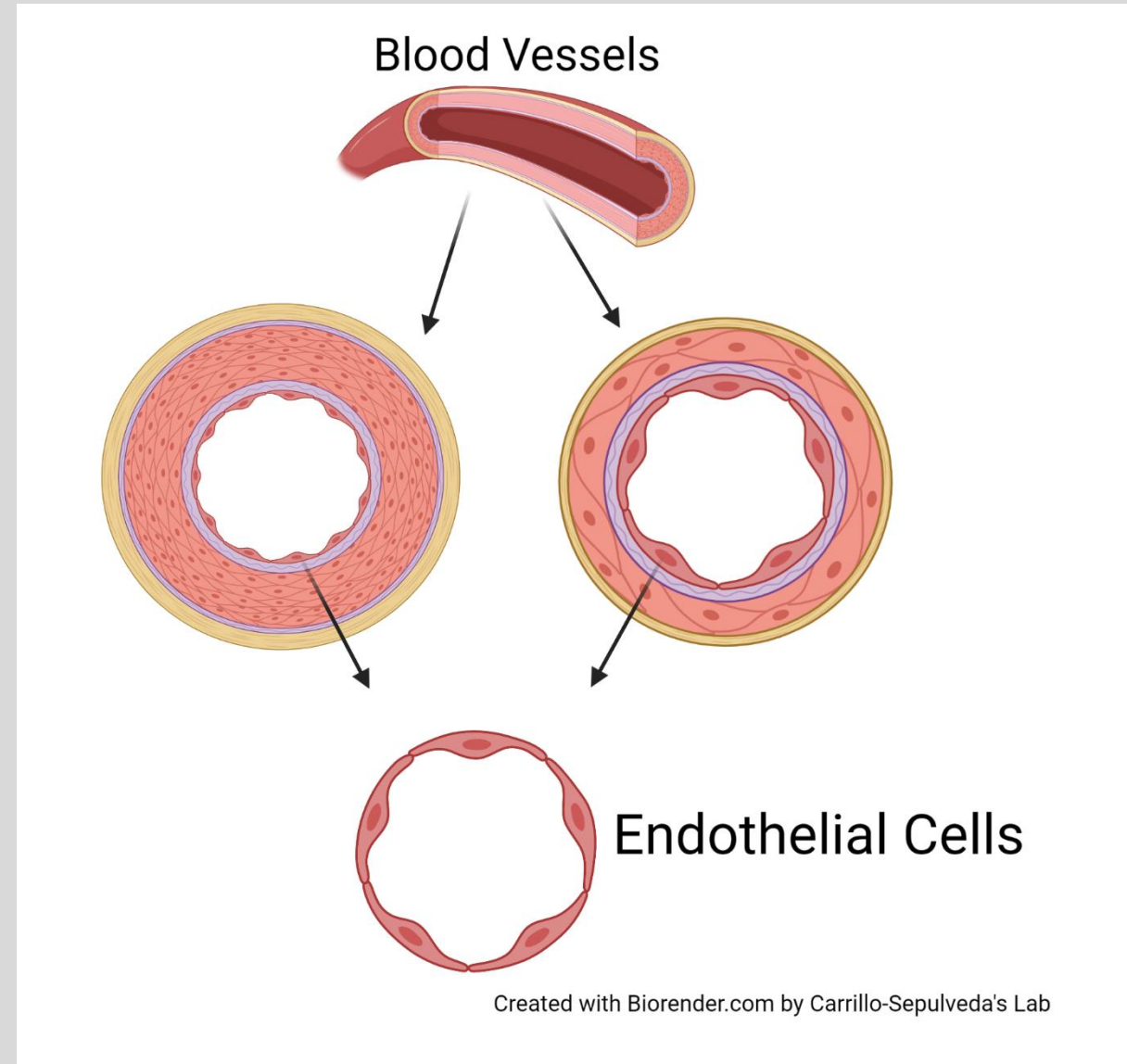
- Endothelial cells produce many factors to maintain homeostasis
- They release **nitric oxide (NO)** that promotes vasodilation
- They can produce reactive oxygen species (ROS) that causes oxidative stress and vascular damage
- They produce thromboxane and prostacyclin that can be important in disease states
- **We will learn the physiological conditions and what happens when endothelial cells become dysfunctional**



Where do you find endothelial cells?

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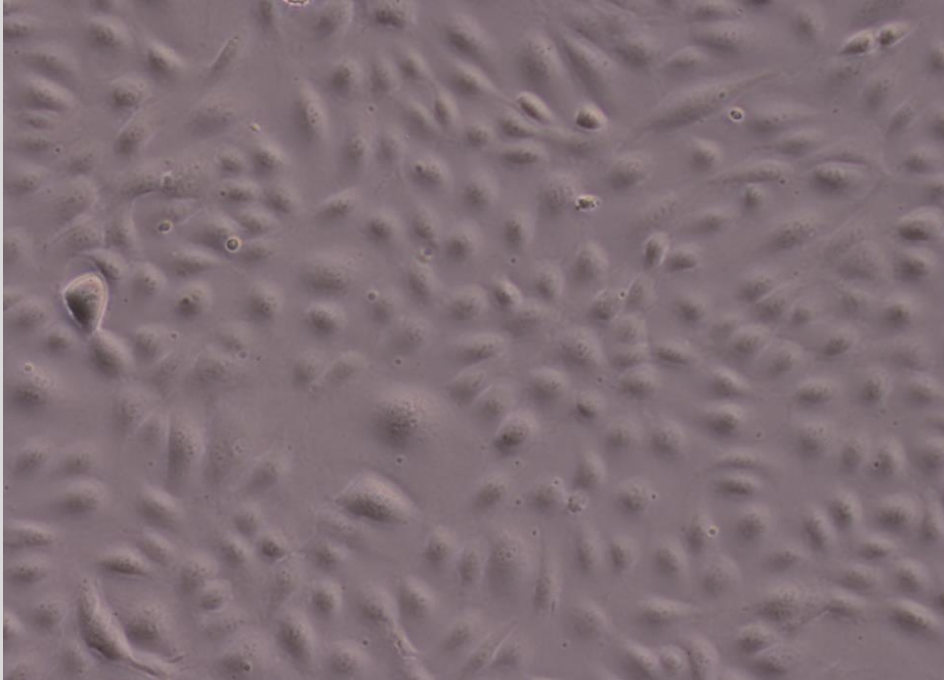
- Endothelial cells are found in **all** blood vessels (arteries and veins, large and small)
- Veins: thinner, weaker
- Arteries: thicker, stronger
- **Compose the inner layer of vessels (endothelium)**



Vascular Cells

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Endothelial Cells



- Cobblestone Shape
- Only one nucleus
- Grow in **monolayer** (even in culture)

Vascular Smooth Muscle Cells

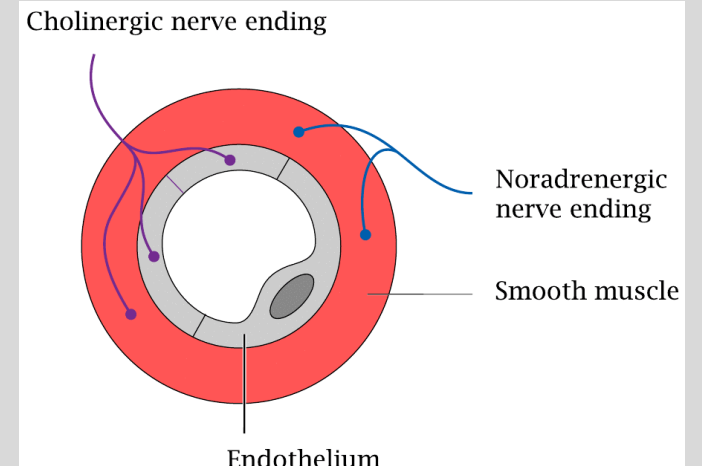
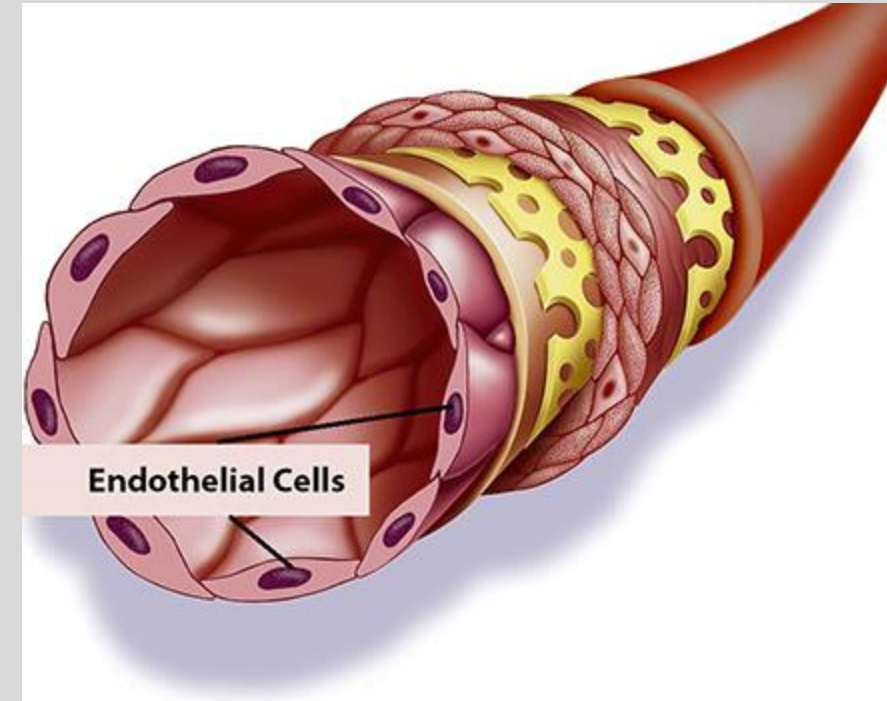


- Spindle Shape
- Can have multiple nuclei
- Grow in **multilayers** (even in culture)

Endothelial Cells

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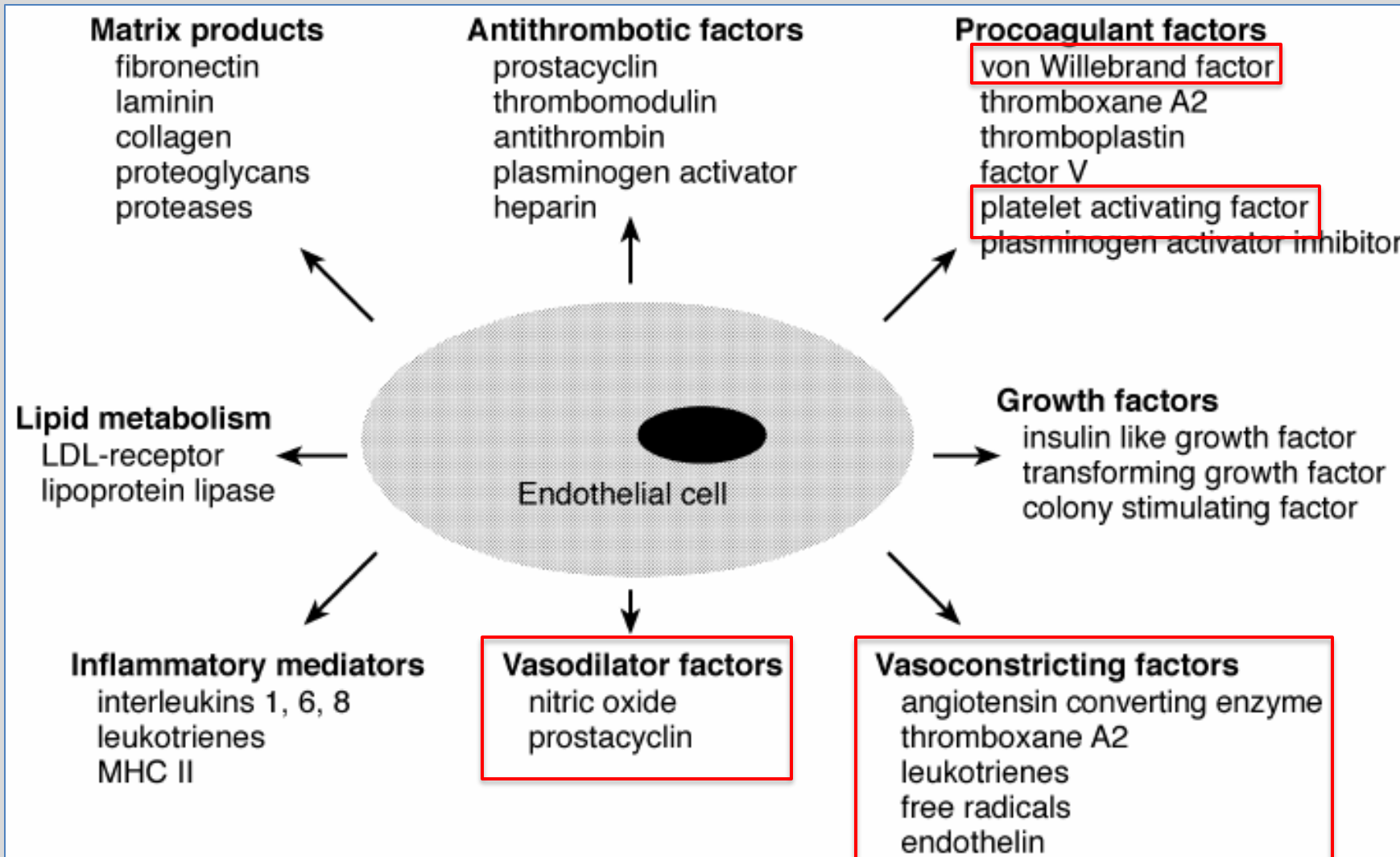
- Primary function of endothelial cells is a **physical barrier**
 - Additional research has discovered a much larger role of endothelial cells in human physiology
- **Endothelium is a monolayer of endothelial cells that lines the blood vessels**
- There are approximately 60 trillion endothelial cells in the human body.
- The **endothelium is innervated by cholinergic nerves and respond to acetylcholine (ACh)**
 - The vascular smooth muscle cell (VSMC) layer is primarily innervated by noradrenergic nerves and responds to epinephrine



Factors secreted by endothelial cells

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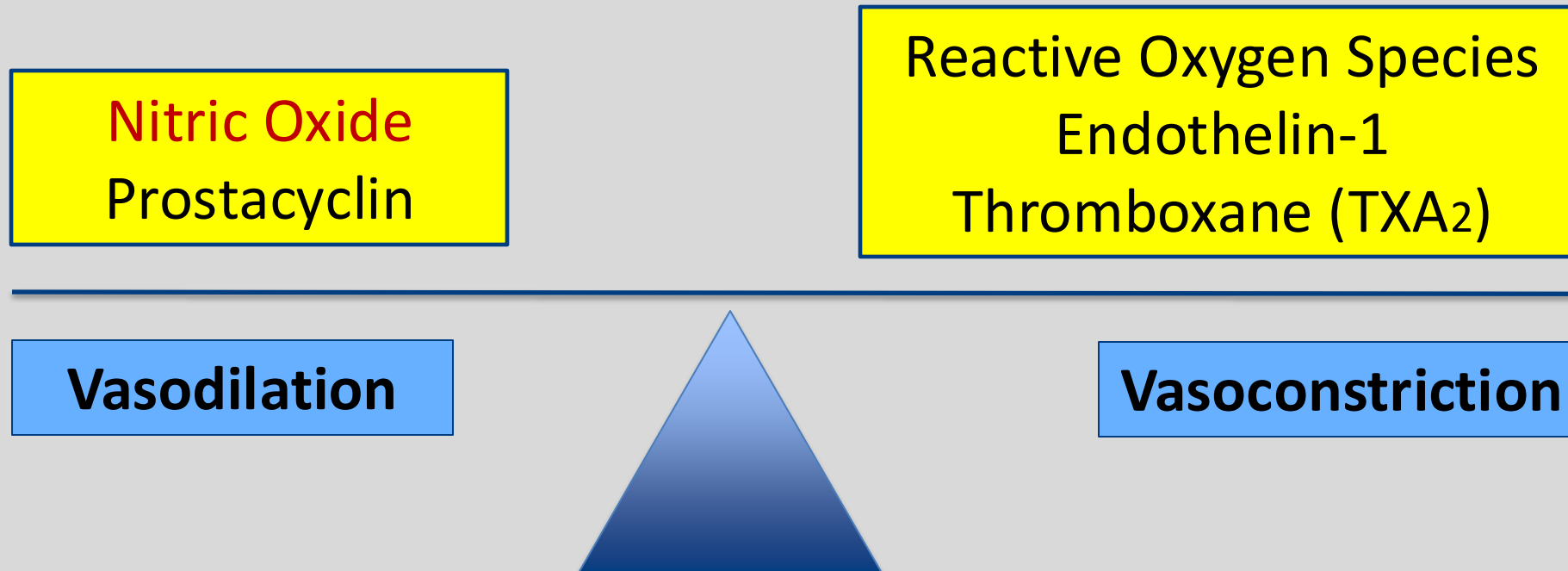


Vasoregulation by Endothelium

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- **Balance between vasodilators and vasoconstrictors maintains physiologic conditions**
- Any **disruption** to the balance will cause conditions with either too much vasodilation or too much vasoconstriction

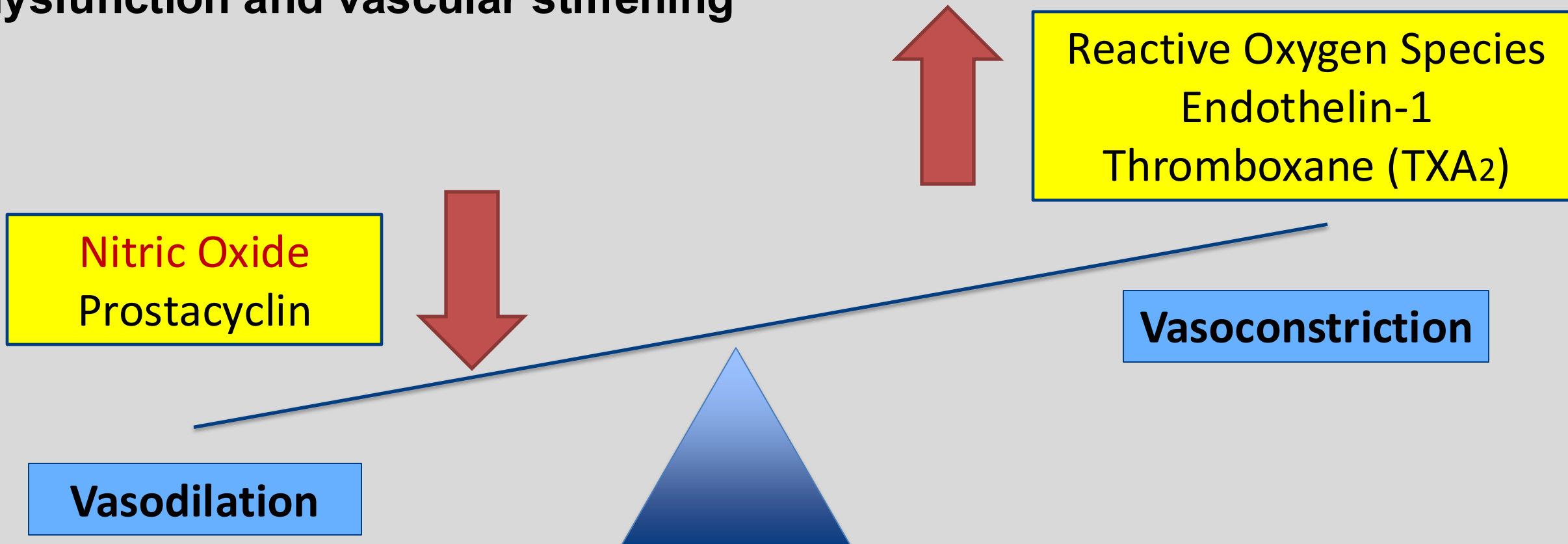


Endothelial Dysfunction

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- Endothelin-1 and TXA_2 induce migration and proliferation of VSMC
- **Nitric Oxide** has anti-proliferative properties
- **Endothelial Dysfunction leads to vascular dysfunction and vascular stiffening**



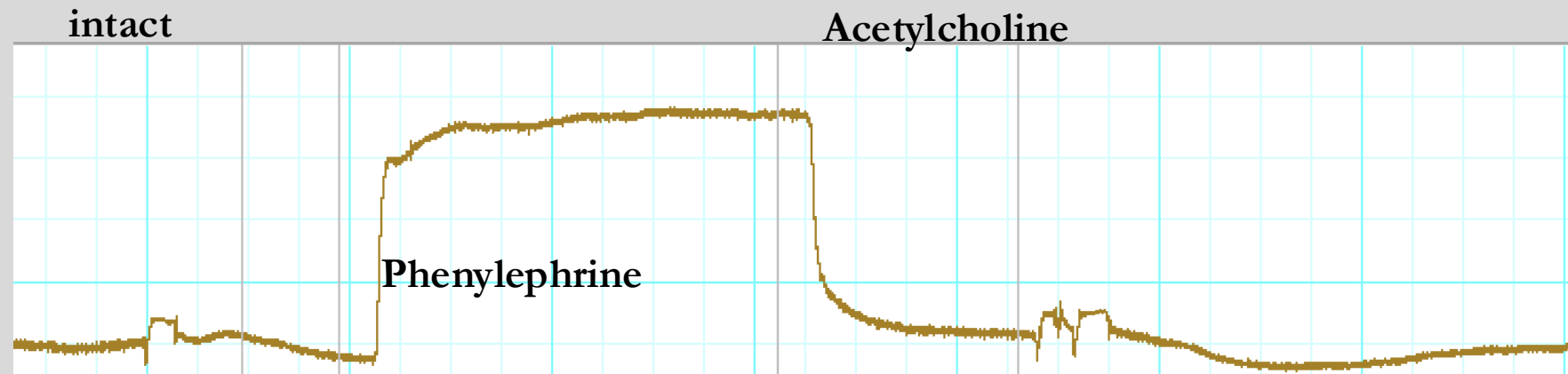
Vasodilation by Endothelium-derived relaxation factor (EDRF)

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- ♥ **Nitric Oxide (NO)** is an EDRF that causes **vasodilation**
- ♥ Acetylcholine (ACh) induces the release of NO leading to relaxation
- ♥ When the vessel is damaged, the integrity of the endothelial layer is compromised.

What will happen to the relaxation curve in the damaged vessel?



Vasodilation by Endothelium-derived relaxation factor (EDRF)

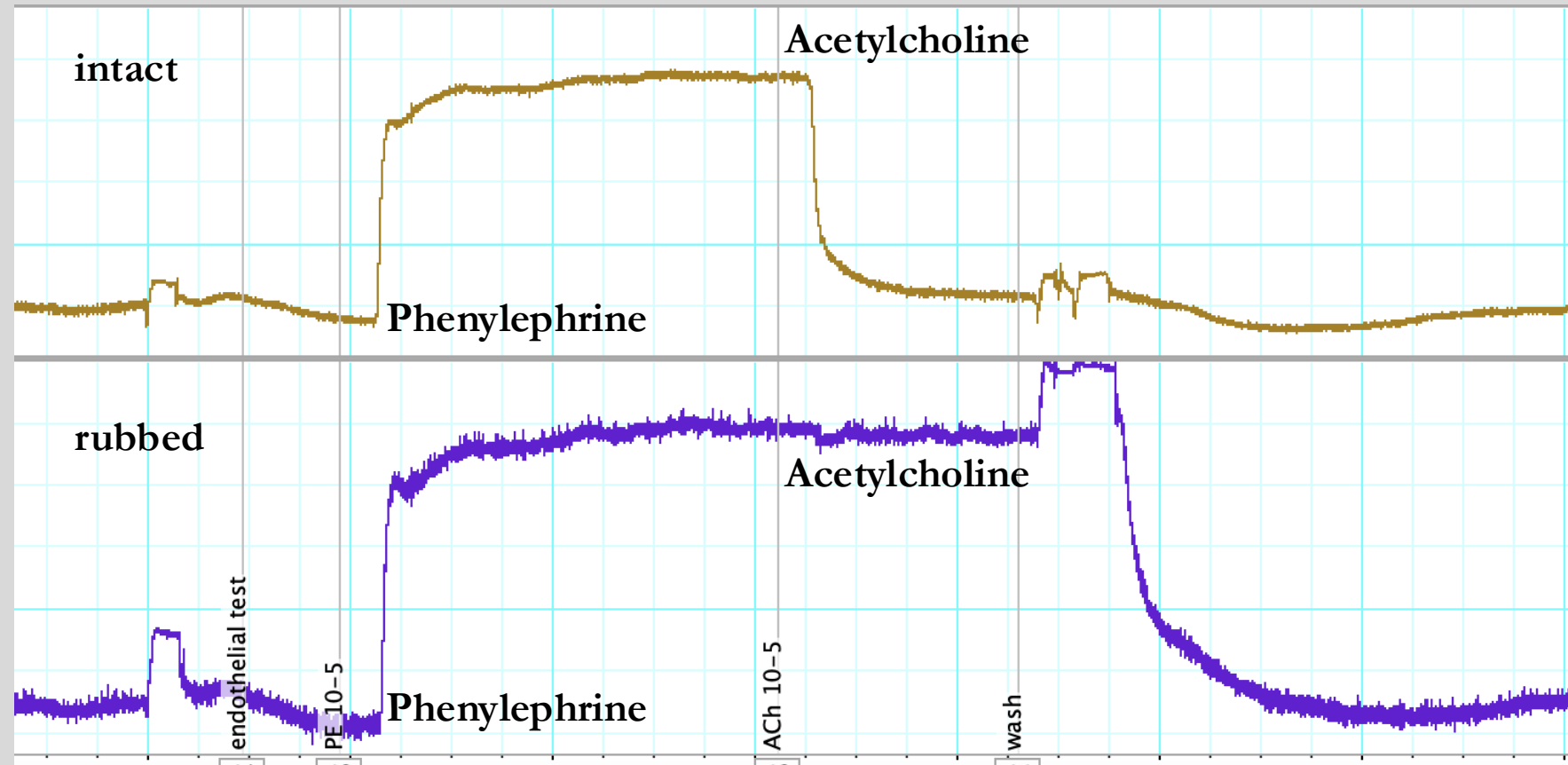
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- ♥ Nitric Oxide (NO) is an EDRF that causes **vasodilation**
- ♥ ACh induces the release of NO leading to relaxation
- ♥ When the vessel is rubbed, the endothelial layer is damaged/removed

What will happen to the relaxation curve in the damaged vessel?

- The vessel will not relax because NO is not being released from endothelial cells



Nitric Oxide Signaling

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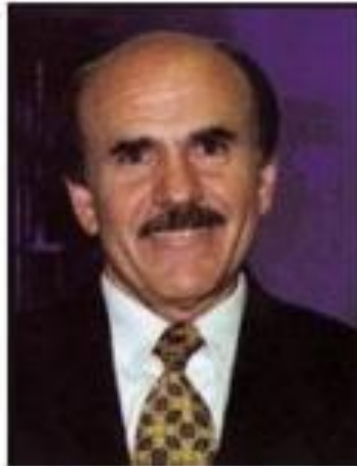
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The Nobel Prize in Physiology or Medicine 1998

The Nobel Assembly at the Karolinska Institute in Stockholm, Sweden, has awarded the Nobel Prize in Physiology or Medicine for 1998 to **Robert F Furchgott**, **Louis J Ignarro** and **Ferid Murad** for their discoveries concerning "the nitric oxide as a signaling molecule in the cardiovascular system".



Robert F Furchgott, born
1916
Dept. of Pharmacology,
SUNY Health Science Center
New York



Louis J Ignarro, born 1941
Dept. of Molecular and
Medical Pharmacology
UCLA School of Medicine
Los Angeles



Ferid Murad, born 1936
Dept. of Integrative Biology
Pharmacology and Physiology
University of Texas Medical
School, Houston

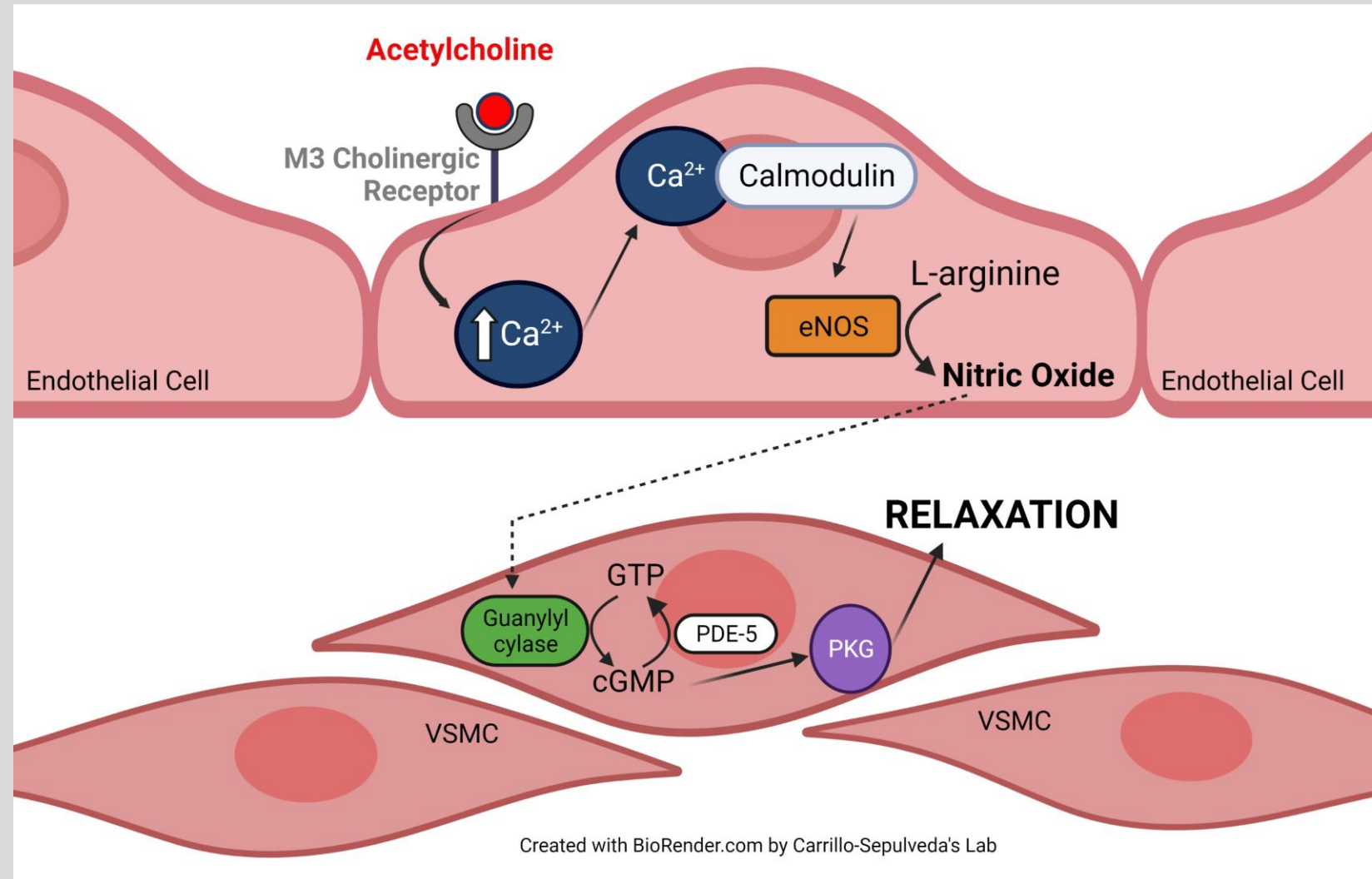
- **NO** is an important signaling molecule in vascular function
- **NO** comes from endothelial cells
- This discovery later went on to assist in the development of multiple drugs such as Viagra

For their discoveries concerning nitric oxide as a signaling molecule in the cardiovascular system

Nitric Oxide Signaling Cascade

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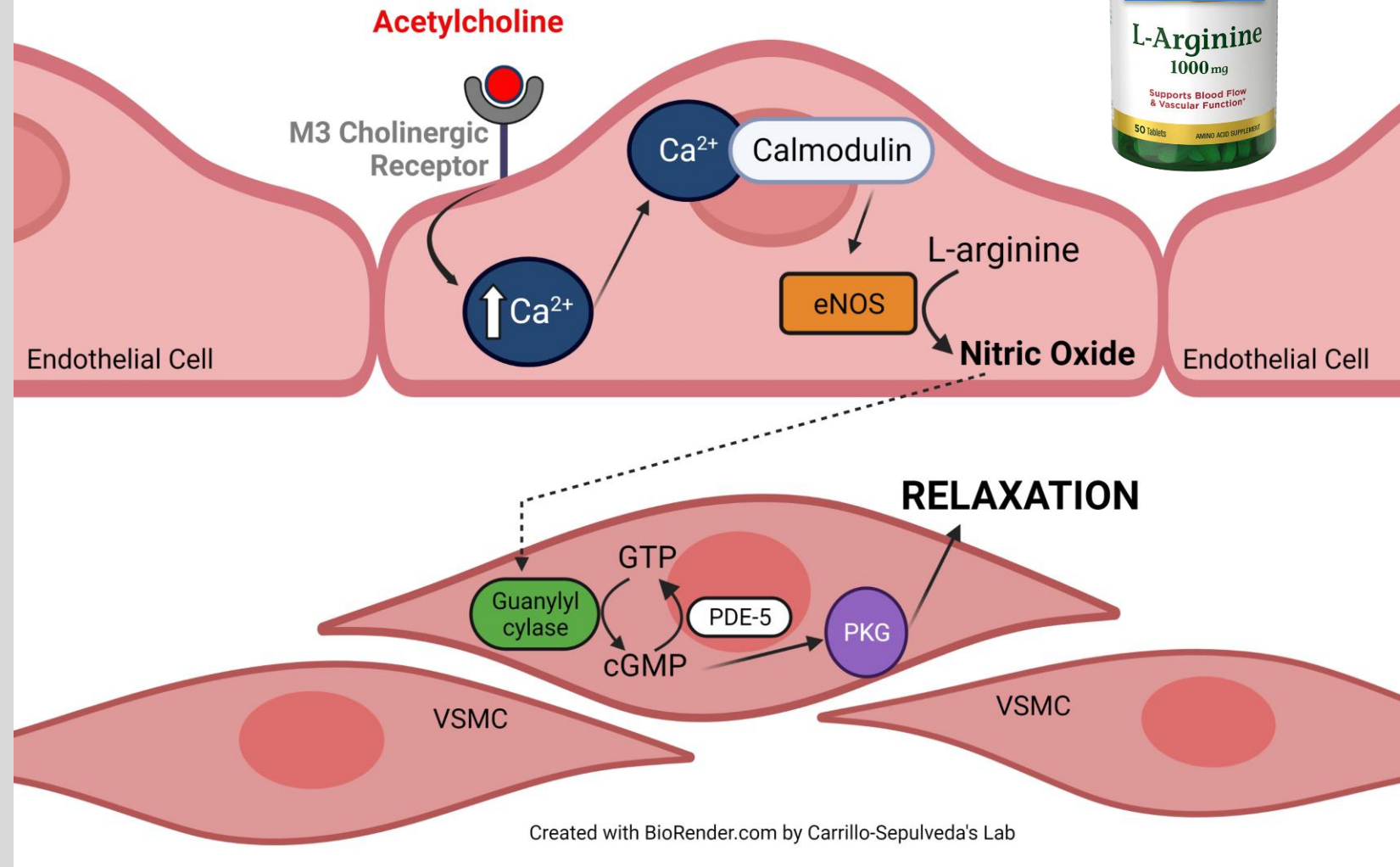


1. **ACh binds the M3 receptor** on endothelial cells and releases **Ca²⁺** into the cell
2. **Ca²⁺** binds to calmodulin, activating endothelial nitric oxide synthase (eNOS)
3. **eNOS converts L-arginine into NO**, which diffuses into VSMCs
4. **NO activates GC to convert GTP into cGMP**
5. **cGMP activates PKG to cause relaxation**

Nitric Oxide Signaling Cascade

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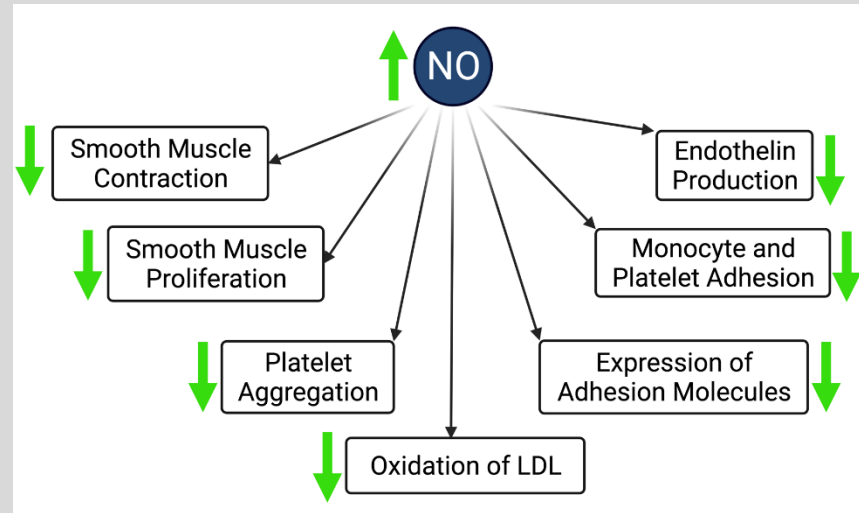
- **PDE-5 degrades cGMP** back into GTP to reverse vasodilation
 - Viagra is a PDE-5 inhibitor
 - **Viagra keeps cGMP levels high leading to increased vasodilation**
- **Shear stress will also activate this pathway**, releasing NO and causing vasodilation

Nitric Oxide Signaling Cascade

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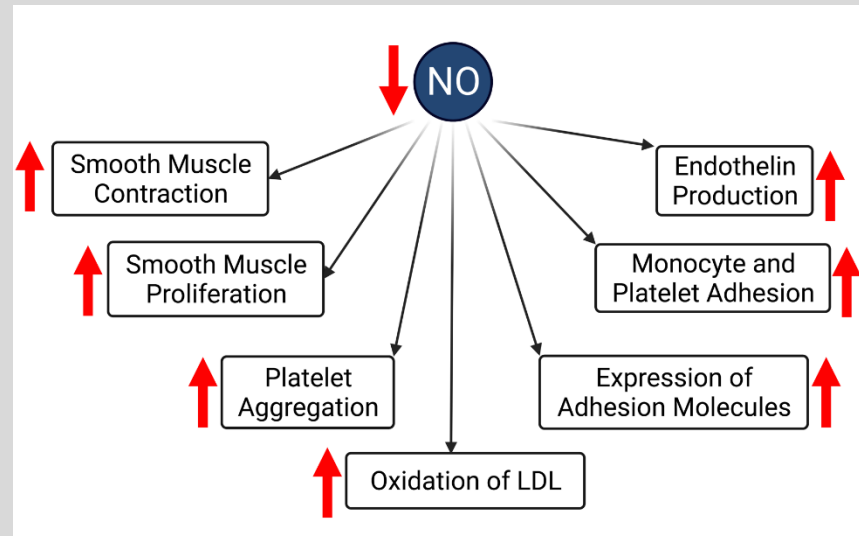
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Normal Endothelial Cells



- NO production causes more than just vasodilation

Dysfunctional Endothelial Cells

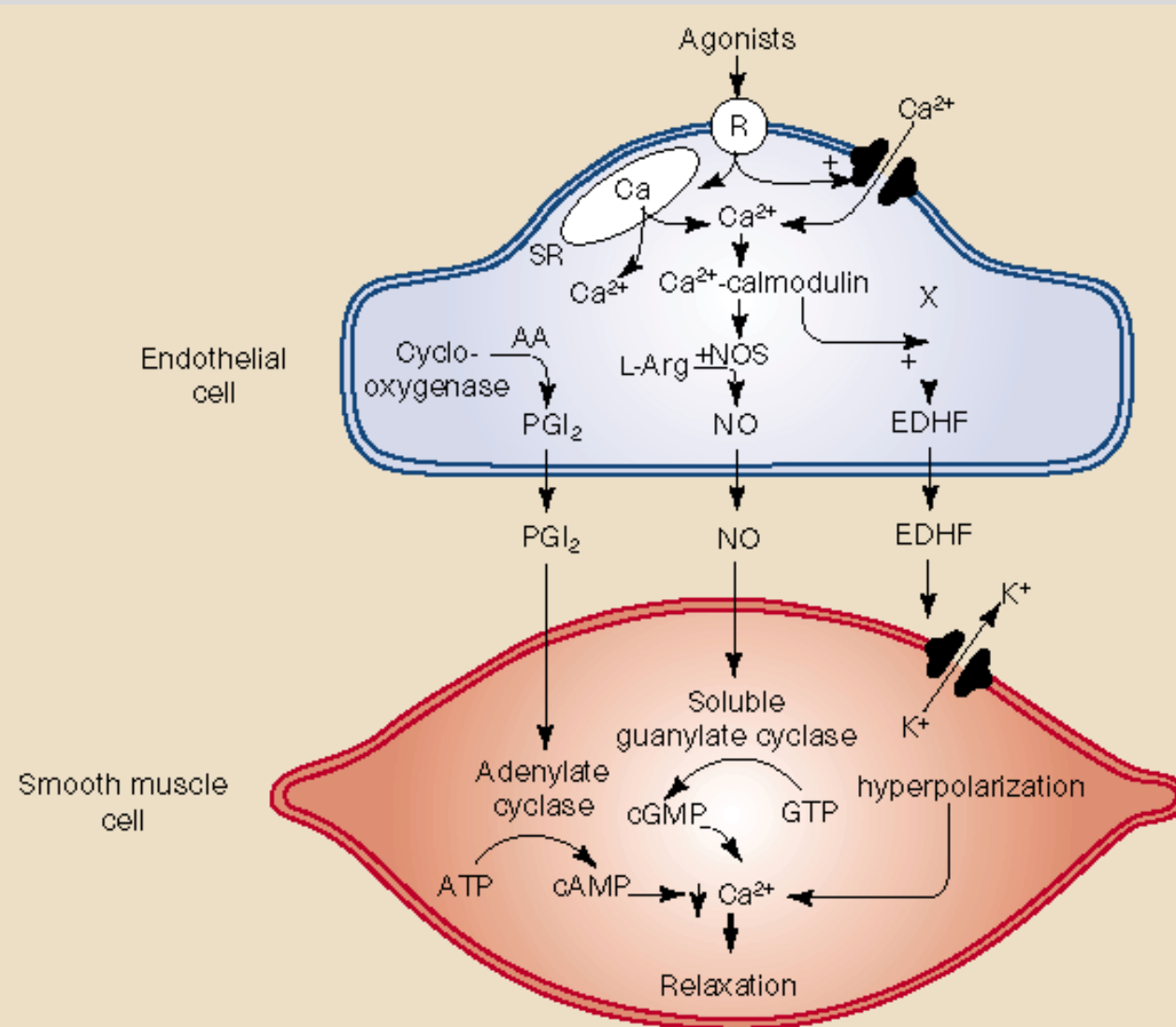


- **Low NO production** causes vasoconstriction, **smooth muscle proliferation**, etc.
- Seen in many disease states

Mechanisms by which endothelial cells causes relaxation

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Practice Question

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A 25-year-old female patient presents to the emergency department with sudden onset difficulty breathing, itchy rash and nausea after dining at a seafood restaurant. Pulse is 125/min, respirations are 24/min, oxygen saturation is 97% and blood pressure is 82/68 mmHg. Skin exam reveals erythematous patches and wheals and her lips are slightly swollen, without swelling of the tongue or uvula. What is the most appropriate next step in management?

- a. IV steroid administration
- b. Nebulized albuterol administration
- c. Endotracheal intubation
- d. IM epinephrine administration
- e. IV diphenhydramine administration

Practice Question

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- d. IM epinephrine administration**
- e. IV diphenhydramine administration

Severe Allergic Reactions

- Allergen exposure leads to increased levels of bradykinin (inflammation) and histamine (from mast cells)
- **Bradykinin and histamine cause vasodilation by increasing NO production in endothelial cells**
- Treatment with epinephrine reverses the reaction by causing vasoconstriction
- **Epinephrine activates alpha adrenergic receptors in the smooth muscle layer**

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Diseases Associated with Endothelial Dysfunction

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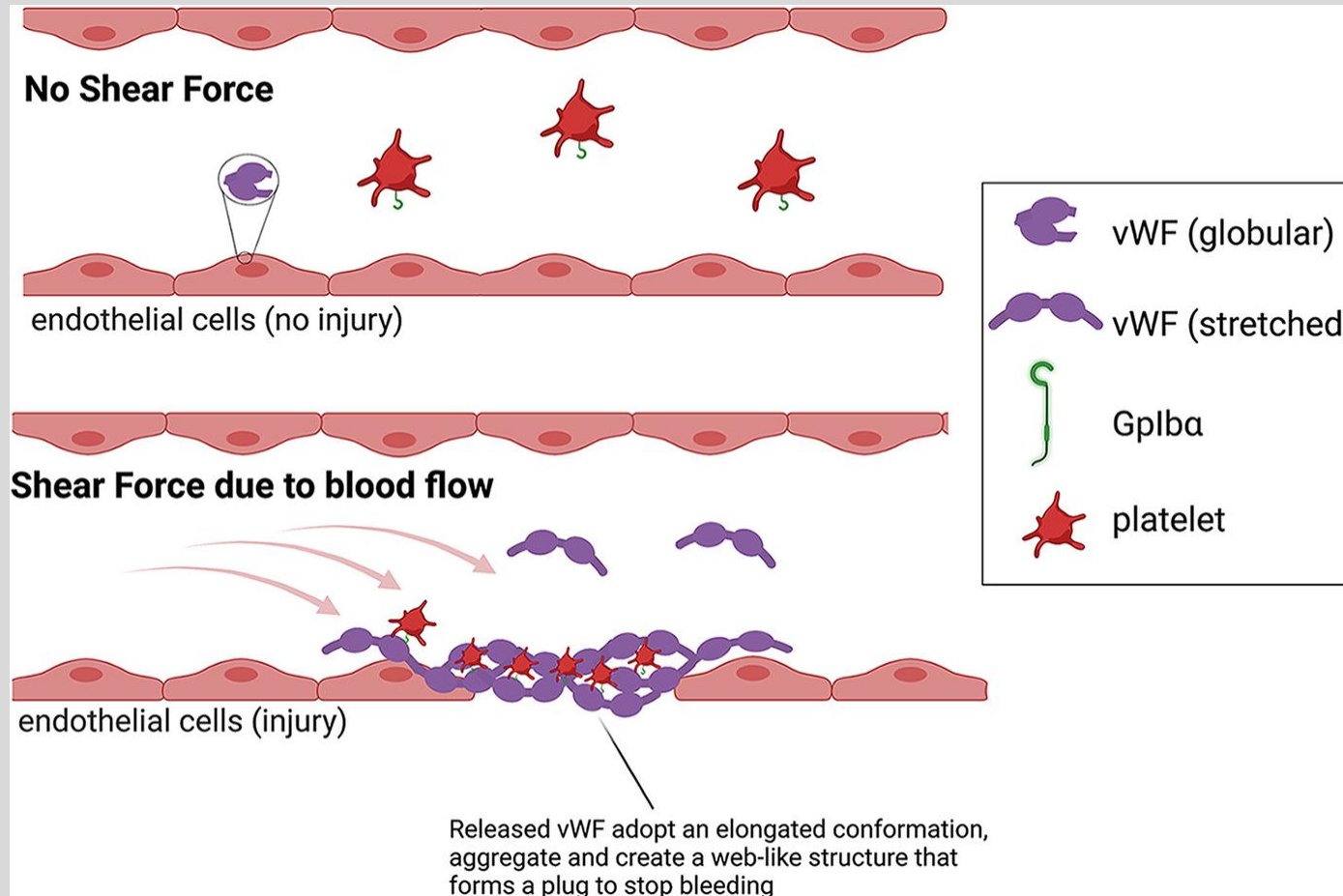
- Hemolytic-Uremic Syndrome
- **Von Willebrand Disease**
- Hypertensive Retinopathy
- **Hypertension**
- Preeclampsia
- Aging
- **Obesity**
- **Deep Vein Thrombosis**
- **Pulmonary Hypertension**
- **Type II Diabetes Mellitus**
- **SARS-COV-2 – COVID-19**

Von-Willebrand Disease

- Von Willebrand Factor (vWF) is a coagulation factor only released by endothelial cells.
- Von Willebrand disease is an autosomal dominant genetic disease characterized by decreased vWF.
- vWF binds to factor VIII and platelets to assist in the formation of a platelet plug

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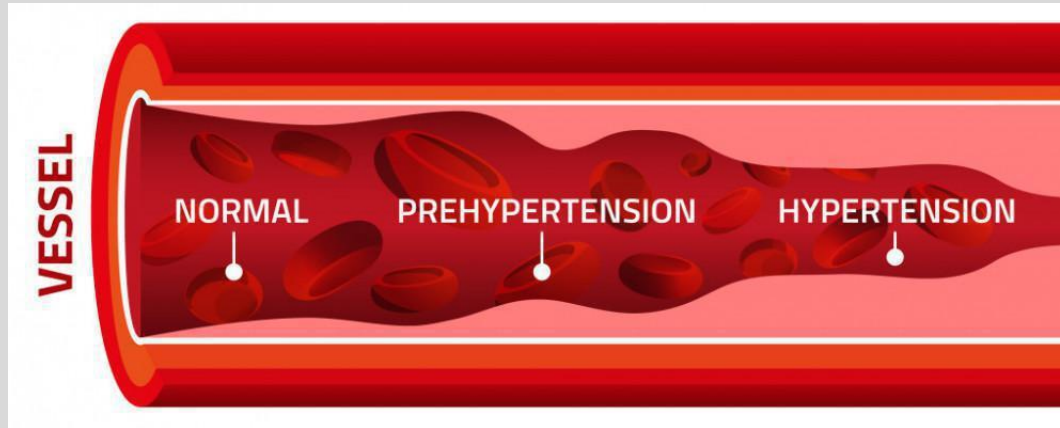


- Treatment is **Desmopressin**, which releases vWF and Factor VIII from endothelial cells

Hypertension

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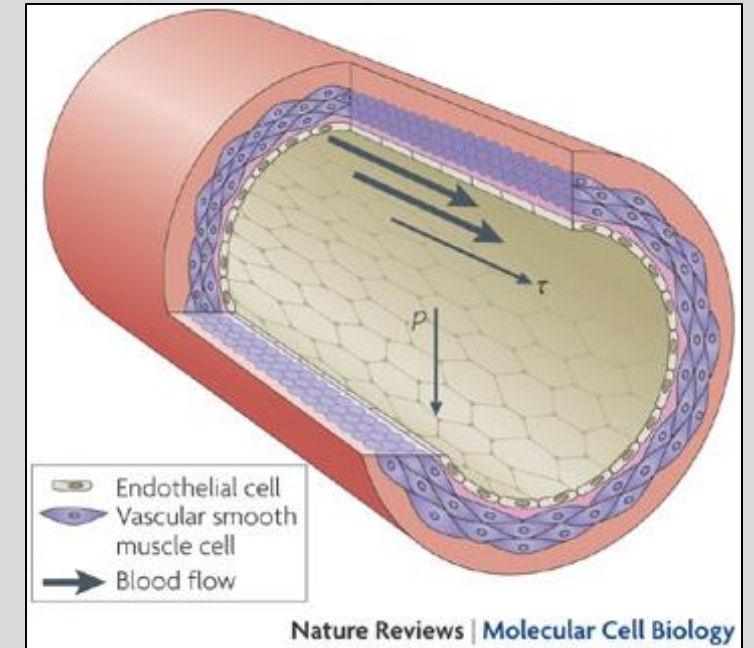
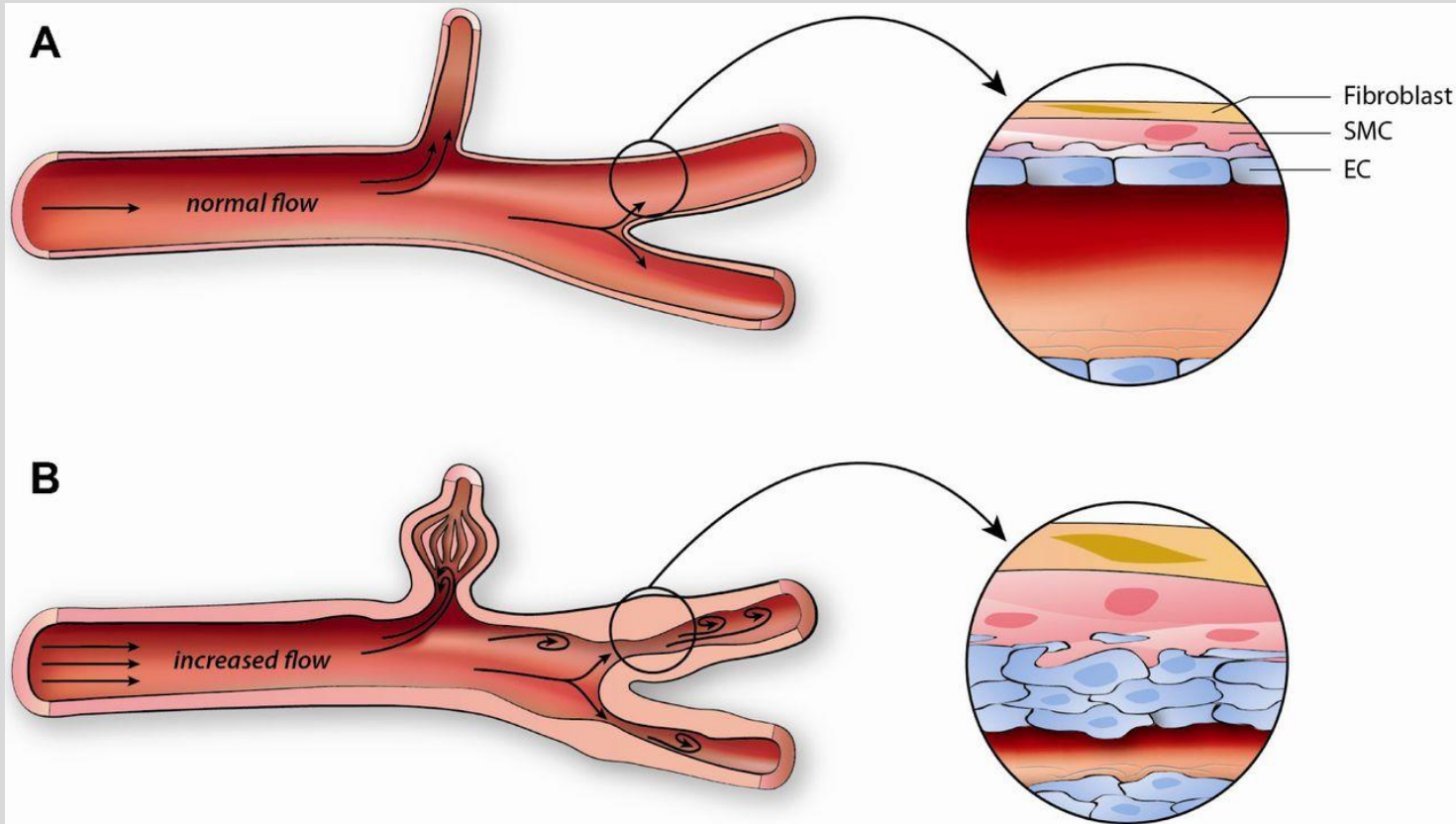
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- ♥ Endothelial dysfunction causes hypertension (HTN)
- ♥ **Decreased levels of NO causes increased vasoconstrictor tone leading to increased BP**
- ♥ HTN can also cause additional endothelial damage by shear stress forces

Shear Stress

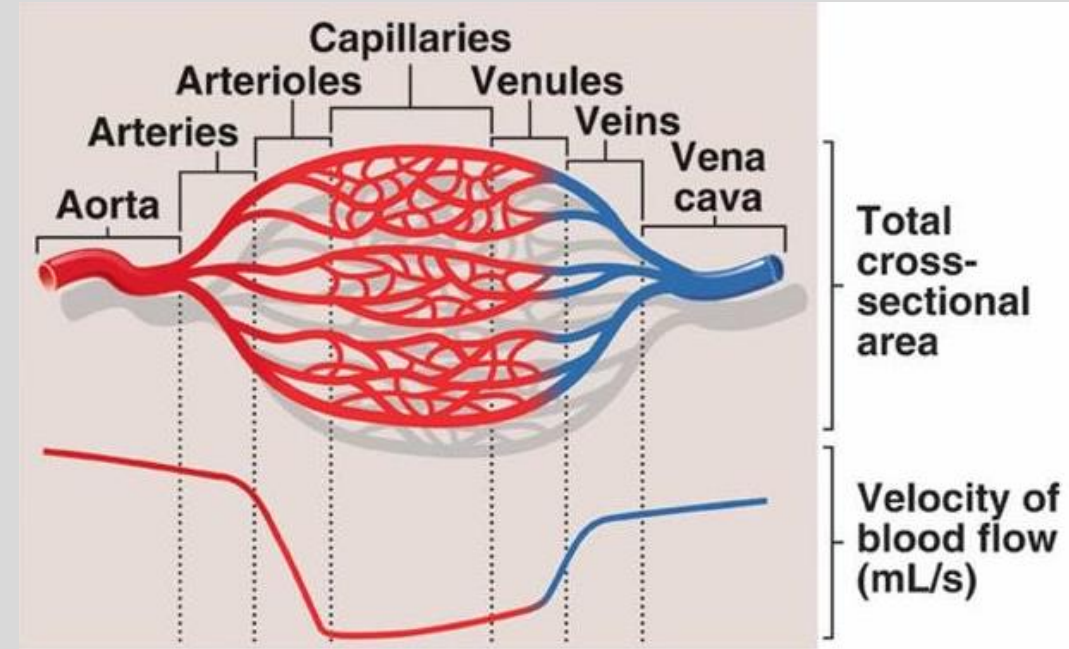
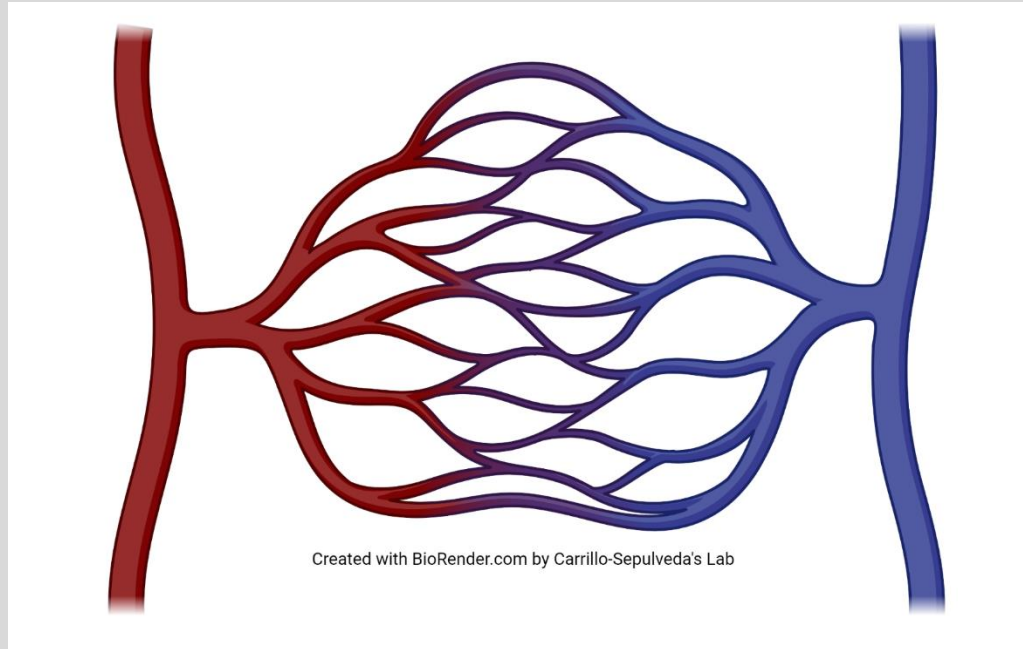
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- **Too much shear stress will damage the endothelial layer**
 - Common at bifurcations causing turbulent blood flow
- **Damage to endothelial cells is a common finding in many diseases**

Blood Flow Dynamics across the Vascular System

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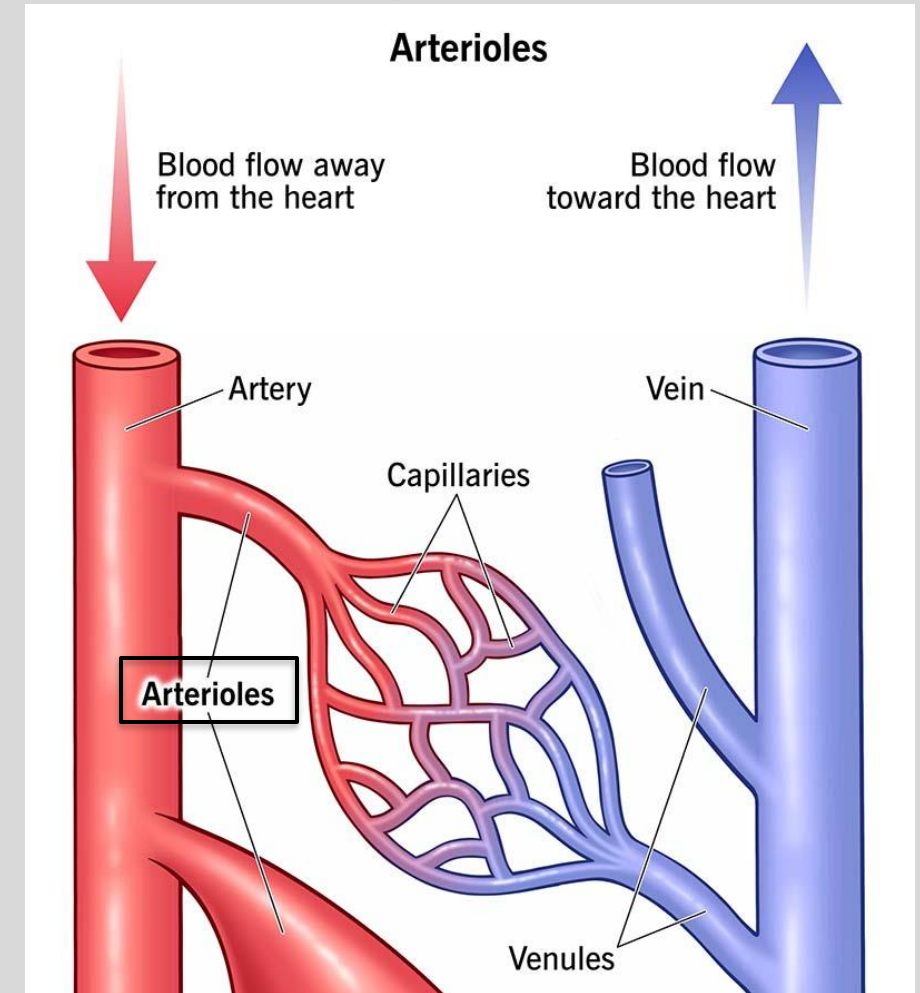


Blood flow velocity is inversely proportional to the **total** cross-sectional area

- Aorta has the smallest total CSA and the highest blood flow velocity
- Capillaries have the largest total CSA and lowest blood flow velocity
- Aorta: conducting vessel, meant to propel blood to the rest of the body
- Capillaries: location of nutrient and oxygen exchange

Total Peripheral Resistance

- ♥ **Arterioles account for most of TPR** Ex: mesenteric arteries
- ♥ TPR plays an important role in blood pressure determination
 - $BP = CO \times \underline{TPR}$
- ♥ **Resistance is inversely proportional to blood flow**



Mesenteric Artery Vascular Reactivity

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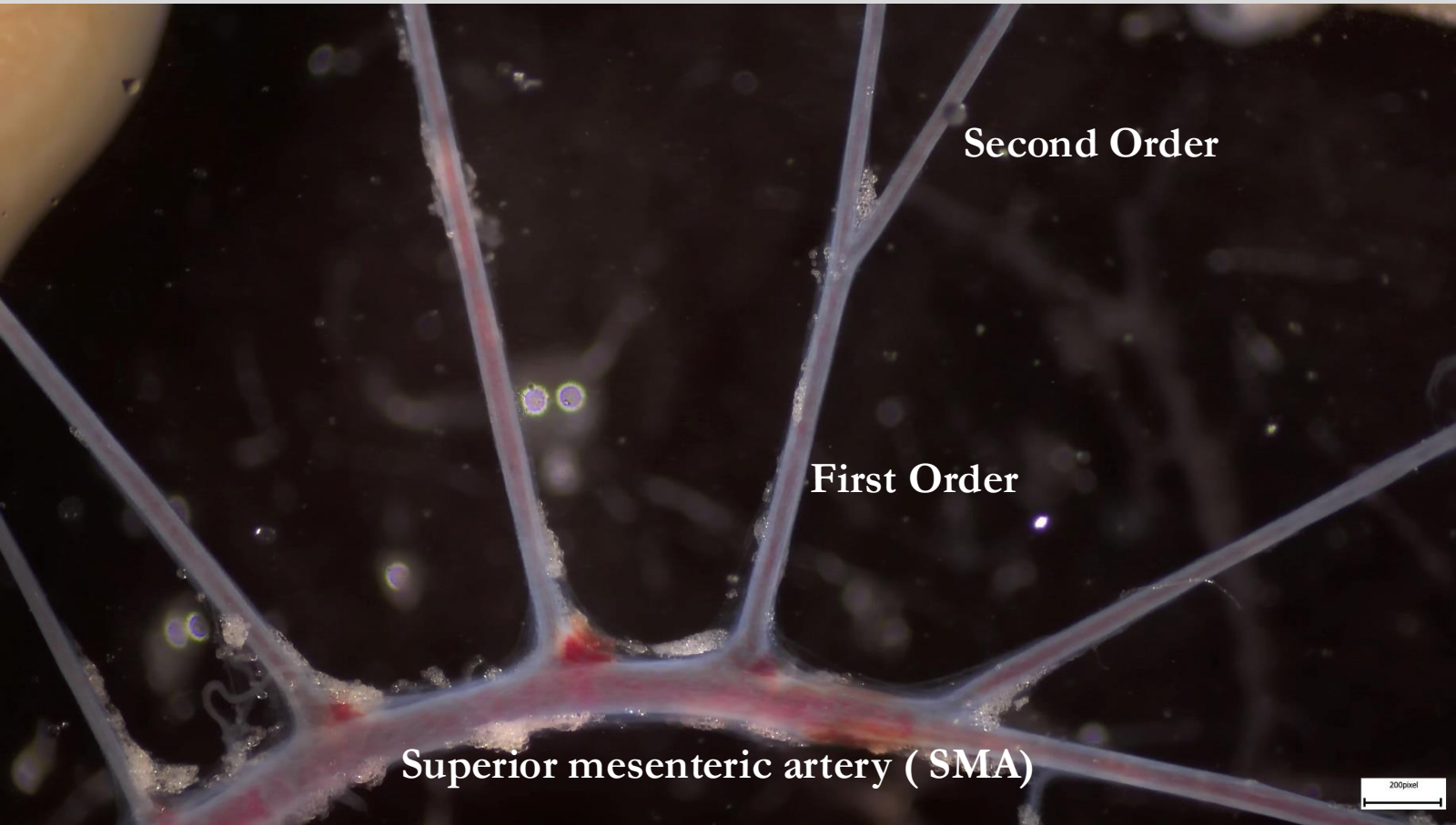


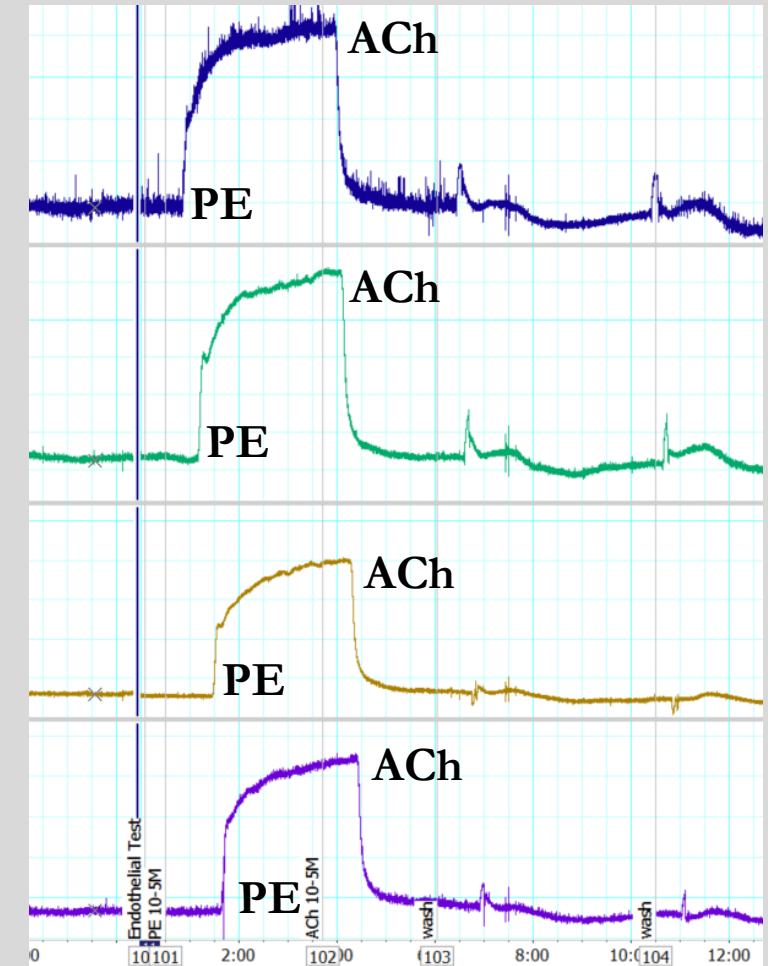
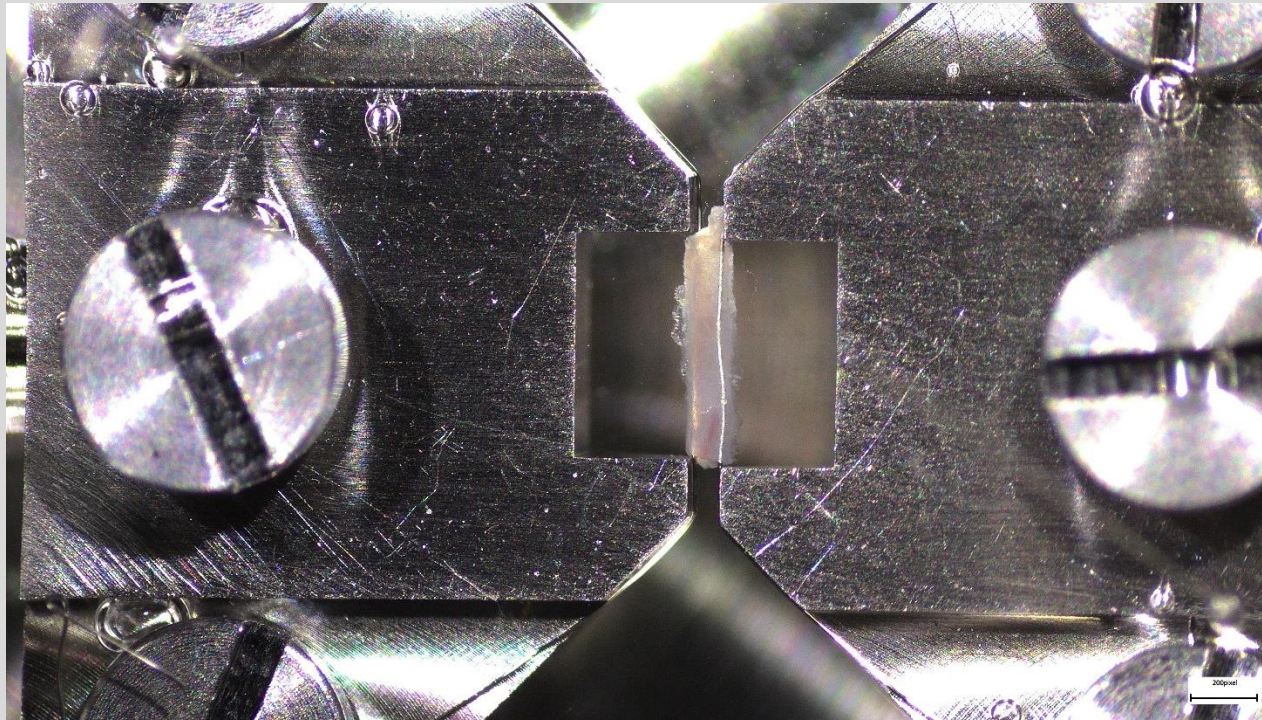
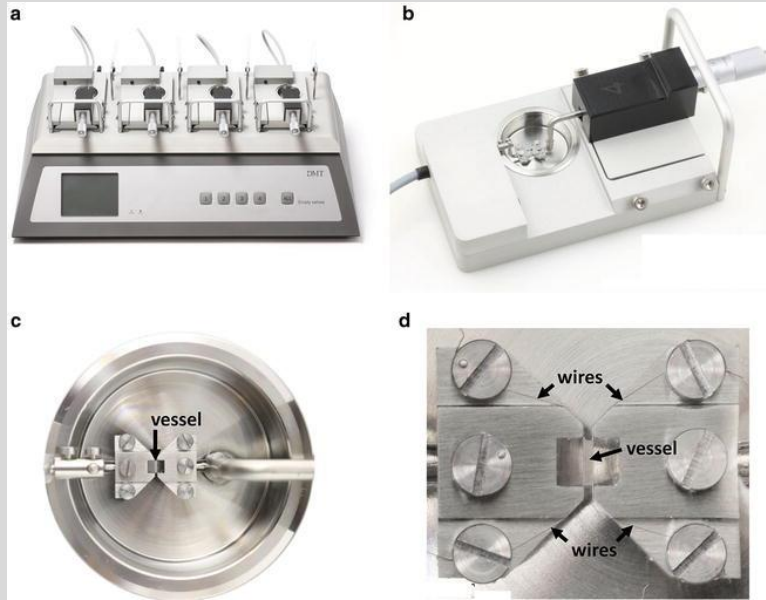
Image from NYITCOM – Dr.Sepulveda's Laboratory

- To assess vascular function, we use 1st order mesenteric arteries because they are resistant vessels (arterioles) responsible for the majority of BP control

Mesenteric Artery Vascular Reactivity

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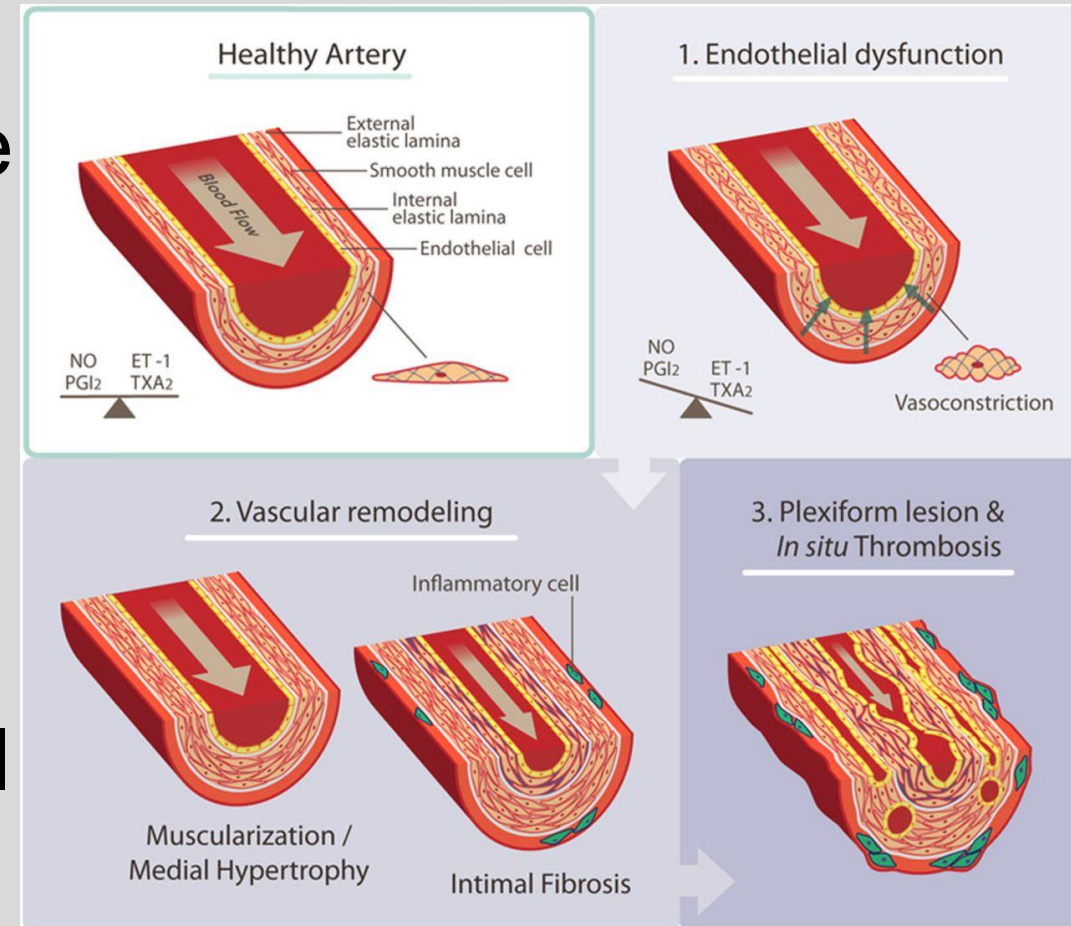
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Pulmonary Arterial Hypertension

- Elevated mean pulmonary artery pressure (>20 mmHg) at rest
- PAH is associated with endothelial dysfunction in pulmonary vasculature leading to **increased vasoconstrictors (endothelin)** and **decreased vasodilators (NO and prostacyclins)**
 - Treat with **bosentan** (endothelin-1 receptor antagonists) or **Sildenafil** (Viagra; PDE-5 inhibitor)

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**Endothelial
Cells**

Endothelium

Endothelin

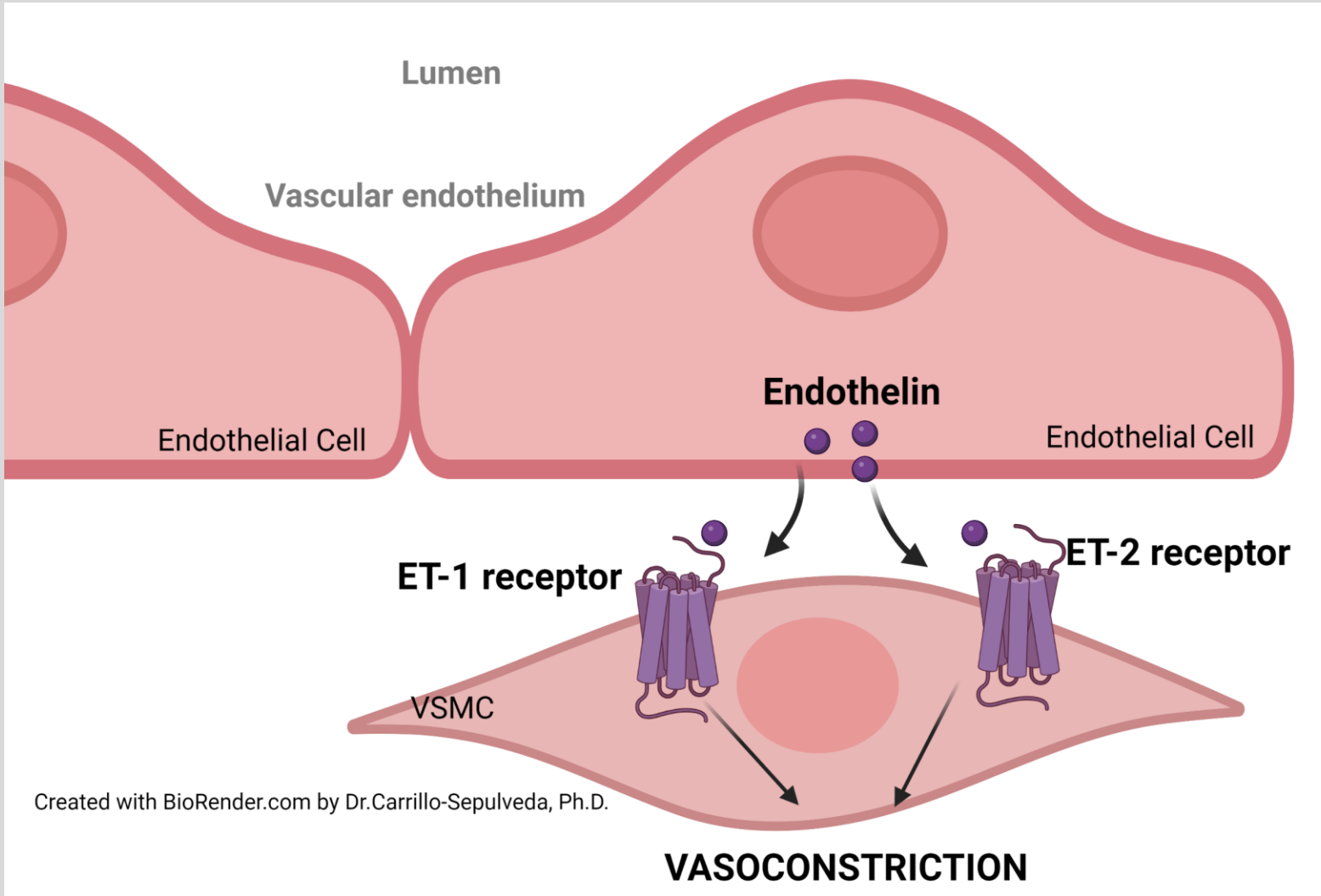
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Type II Diabetes Mellitus

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- ♥ Endothelial Dysfunction
- ♥ Less NO production
- ♥ Endothelin-1 production is increased in T2DM
- ♥ **Endothelin-1 activates ET_A and ET_B on VSMC to cause vasoconstriction and smooth muscle proliferation**

Obesity

- Many believed that all endothelial damage came from inside the vessel (“inside-out” theory of damage)
- There is now research supporting an **“outside-in” theory of endothelial damage**
- The fat surrounding the vessels (perivascular adipose tissue) can become dysfunctional and release factors that contribute to endothelial dysfunction and subsequent HTN
- More research is needed to determine the mechanisms involved

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Lean

Obese



Image from NYITCOM – Sepulveda's Laboratory

Deep Vein Thrombosis

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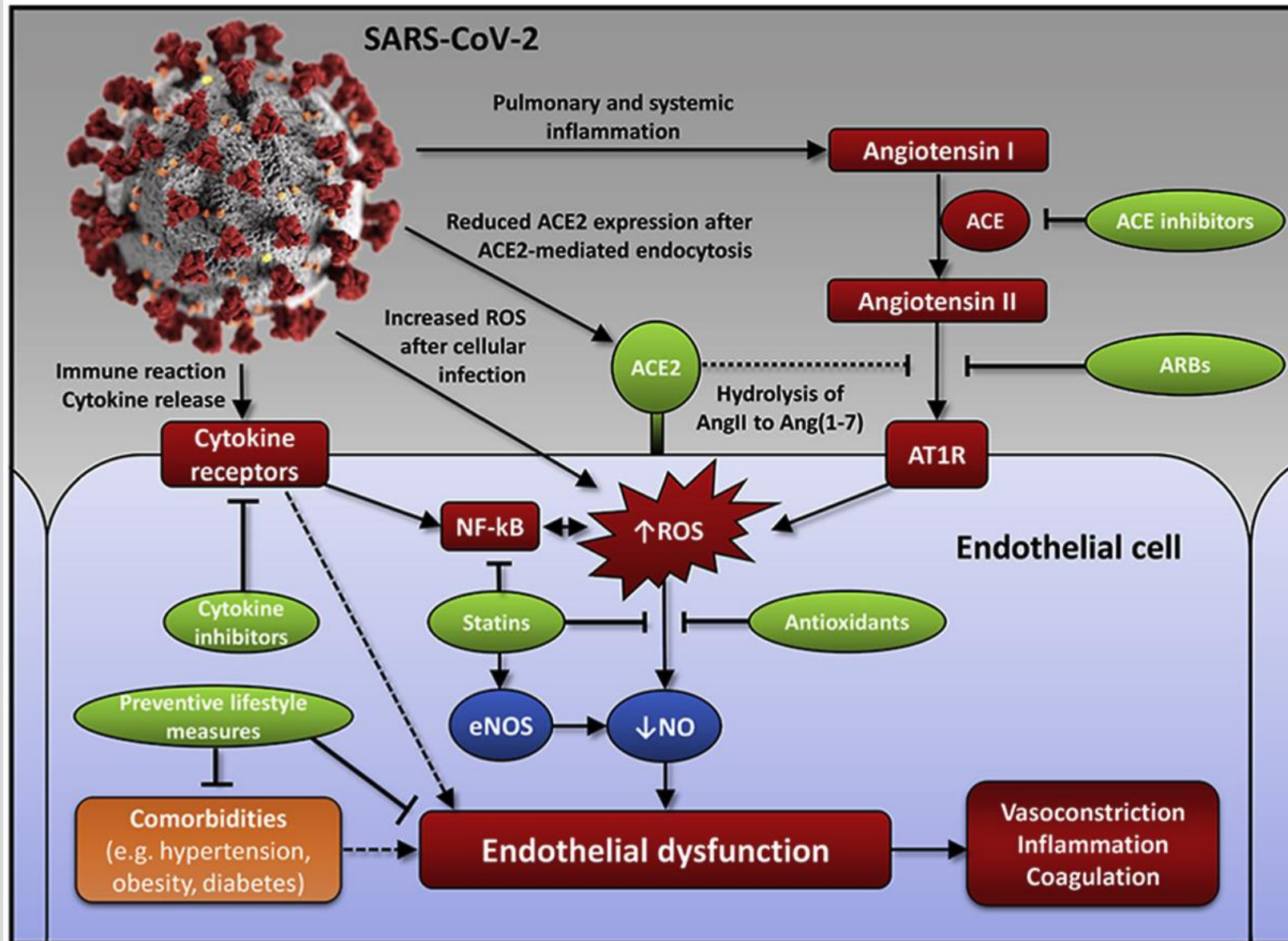
- Endothelial dysfunction predisposes patients to DVT development
- Clinically presents as swelling, redness, warmth and pain
- **Virchow triad**
 - Stasis
 - Hypercoagulability (defect in coagulation cascade factors)
 - **Endothelial Damage (triggers clotting cascade)**
- Diagnose with compression ultrasound with doppler



Endothelial dysfunction and COVID-19

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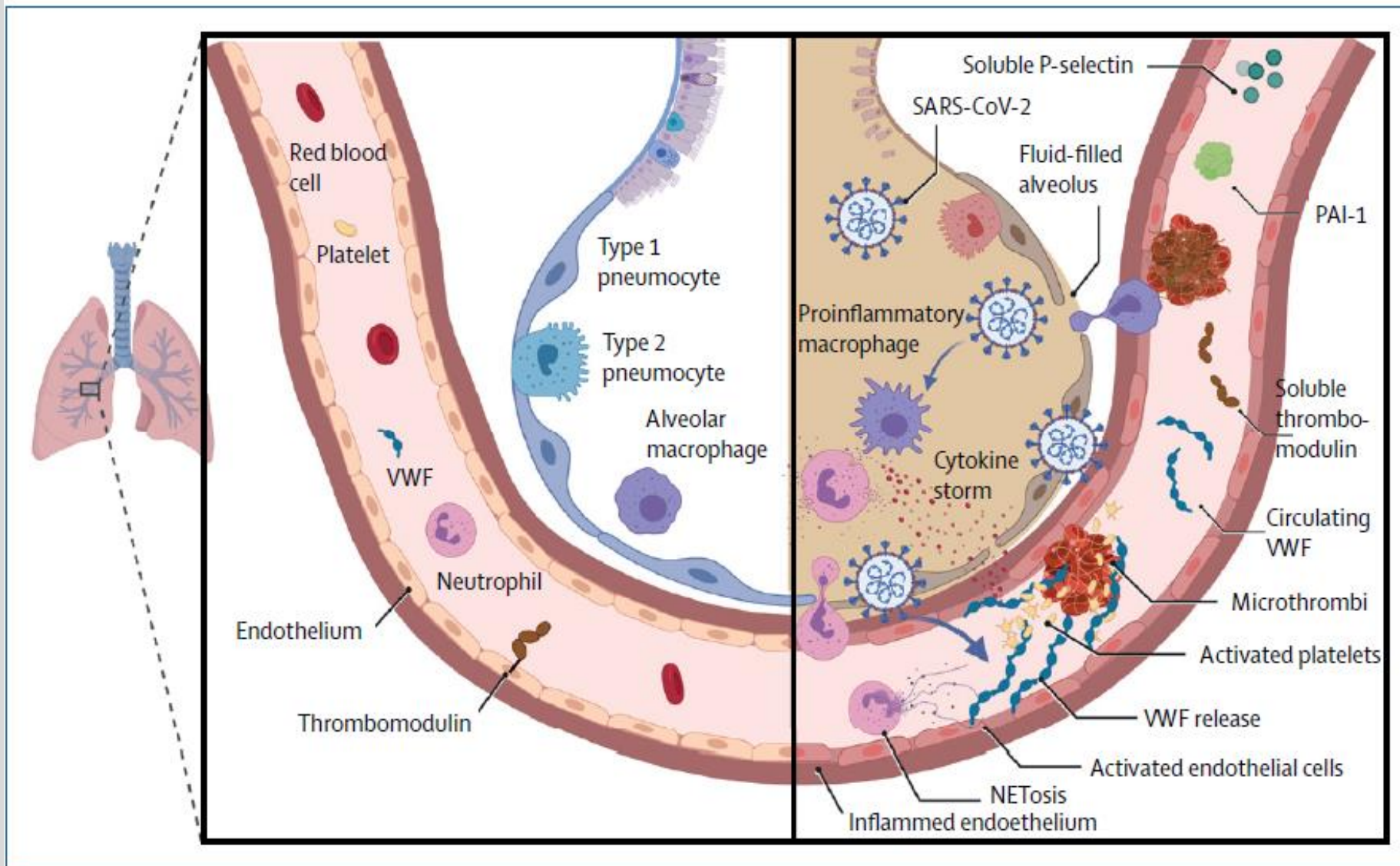
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Endotheliopathy in COVID-19 coagulopathy

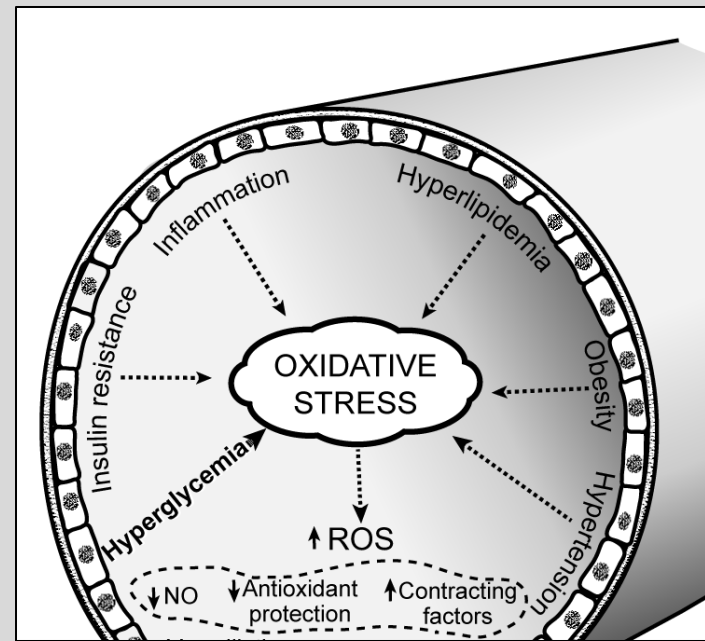
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Treatment Options

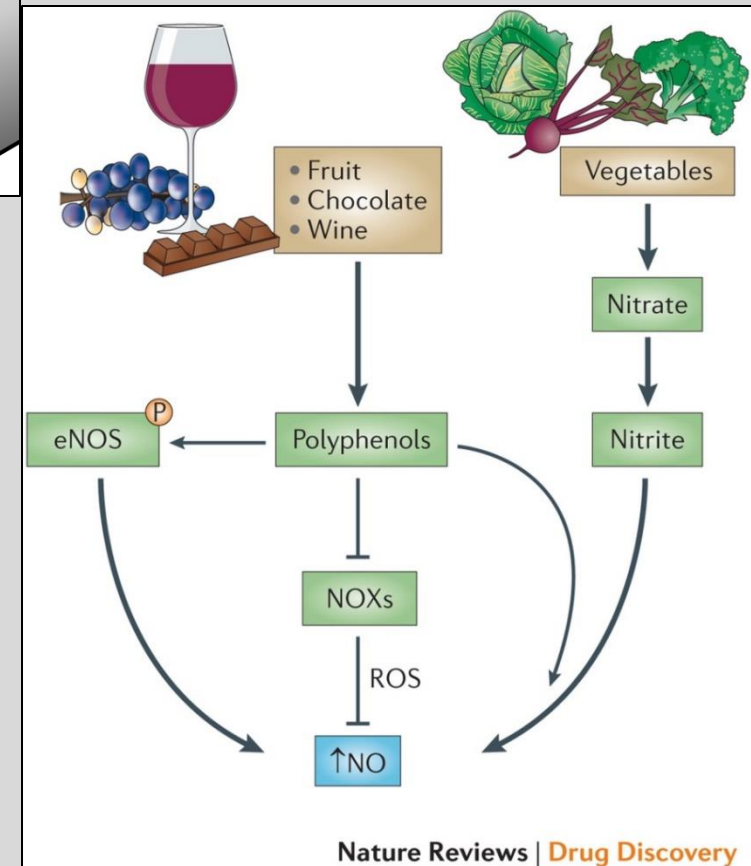
- ♥ Endothelial dysfunction leads to **increased production of ROS leading to oxidative stress**
- ♥ Oxidative stress is commonly seen in diabetes, HTN, inflammation and obesity



- ♥ Oxidative stress results in **increased vasoconstriction (less vasodilation)**
- ♥ Possible treatment is to consume **anti-oxidant foods and medications to prevent the production of ROS**
- ♥ These worked well in culture and lab studies but failed during human trials
- ♥ **Phenols** are thought to block ROS production by **increasing eNOS and blocking the NOX proteins that produce ROS**

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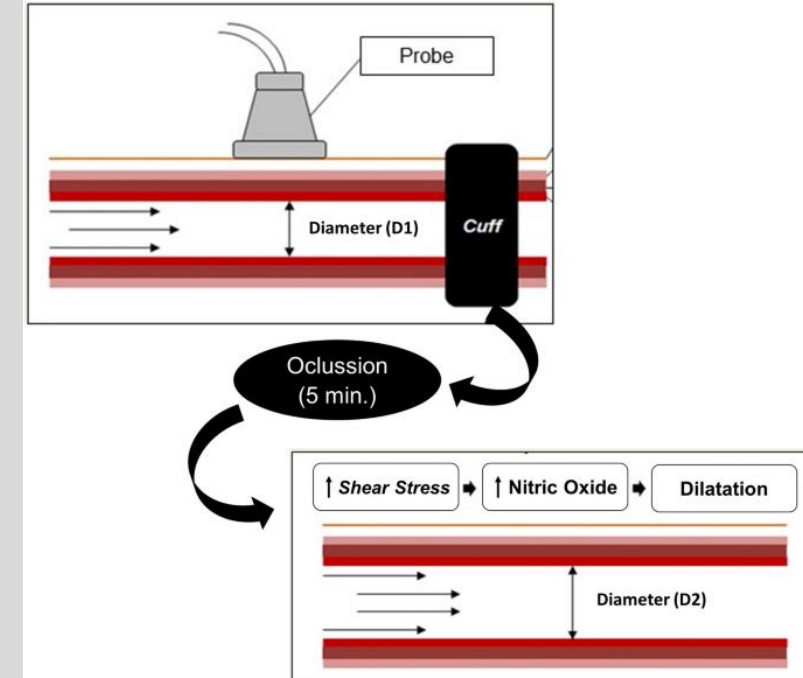
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Clinical Assessment of Endothelial Function

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- ♥ Brachial Artery Flow-mediated dilation (FMD) is used to assess endothelial function in clinical trials
- ♥ In the future, the hope is to use this technique to detect endothelial damage before the development of HTN and prevent progression



Practice Question

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A 51-year-old male patient presents to the emergency department complaining of headache and blurred vision. Physical and lab examination revealed blood pressure of 160/95mmHg, A1C 8%, fasting glucose levels of 240mg/dL. Patient has history of coronary disease. Which of the following most likely explain the cardiovascular complication found in this patient?

- a. Increased production of nitric oxide
- b. Increased production of prostacyclin
- c. Decreased production of reactive oxygen species
- d. Decreased production of endothelin-1
- e. Increased production of reactive oxygen species

Practice Question

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Practice Question

A 63-year-old woman with a history of poorly controlled hypertension and type 2 diabetes is found to have reduced flow-mediated dilation on vascular ultrasound, indicating impaired endothelial function. Which of the following most likely contributes to her elevated blood pressure?

- a. Decreased endothelin-1 production
- b. Increased nitric oxide release
- c. Decreased nitric oxide availability
- d. Increased prostacyclin production
- e. Decreased sympathetic tone

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Summary

- Endothelium is an endocrine organ, releasing multiple factors to maintain balance between vasodilation and vasoconstriction
- Endothelial cells are mononuclear, grow in a monolayer and are innervated by M3 cholinergic neurons
- **Nitric oxide** as a key signaling molecule in the cardiovascular system causing VSMC vasodilation
- Sildenafil (Viagra) promotes vasodilation by blocking cGMP degradation
- Endothelial dysfunction is a hallmark of many cardiovascular diseases including HTN, Pulmonary HTN, Obesity, T2DM and DVT and mostly recently COVID-19
- Endothelial cells are the exclusive producer of vWF and play an important role in VWD pathogenesis

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